

Continuum Memory Architecture (CMA) for Clinical Search: A Simulation-Based Evaluation of Intent-Aware Retrieval Using Complex Public Clinical Data

Abstract

Electronic health record (EHR) chart review is a non-linear, multi-turn task in which clinicians follow diagnostic threads and pivot abruptly between topics. Conventional session-based retrieval embeds prior queries uniformly, so stale context contaminates rankings after topic shifts and increases cognitive load. We evaluate the Continuum Memory Architecture (CMA), a retrieval framework that represents session intent as a trajectory on the Riemannian manifold of symmetric positive definite (SPD) matrices, suppresses stale context with a geodesic-shift-aware interference gate, and forecasts the next latent intent with a Joint Embedding Predictive Architecture (JEPA) to prefetch relevant documents. We conducted a randomized three-arm crossover simulation on clinical vignettes derived from publicly available EHR data (MIMIC-III/IV, MIMIC-IV-Note, MIMIC-CXR, MIMIC-CXR-RRG, MIMIC-IV PPG-ECG, eICU, HiRID), comparing CMA with TF-IDF and BM25 session-based baselines, together with a scaling study from 1 to 10,000 vignettes. Primary outcomes were time-to-correct-information and task completion; secondary outcomes included simulated cognitive load (NASA-TLX), query latency, and query volume. All conditions achieved perfect task completion. CMA reduced time-to-correct-information by 17.0% (98.1 s vs 118.2 s), query latency by 20.8% (141.6 ms vs 178.9 ms), and cognitive load by 18.6% (47.0 vs 57.7) on the benchmark case. Gains were stable across scales: 13.1–14.5% faster time-to-information, 20.8–22.9% lower latency, and 15.7–18.6% lower cognitive load, with statistically significant paired comparisons wherever sample size permitted. CMA’s intent-aware session management improves the speed and efficiency of clinical chart review while preserving task completion, with benefits that generalize from a single-vignette benchmark to large-scale workloads.

Keywords: clinical information retrieval, session-based search, symmetric positive definite manifold, JEPA, geodesic shift, EHR chart review

1. Introduction

Longitudinal electronic health record (EHR) review is a fundamentally multi-turn inference task. A clinician moves among medications, laboratory trends, imaging reports, consult notes, and problem lists, often returning to previously discarded hypotheses as new findings emerge. Each query in this process is not drawn independently from a static information need; it is conditioned on the evolving cognitive and informational context of the reviewer.

A persistent failure mode of conventional session-based retrieval is *latent-context pollution*: the leakage of prior query embeddings, term weights, or implicit feedback into the representation of the current intent. When a clinician pivots abruptly—for example, from a cardiology work-up to renal dosing, or from sepsis management to medication reconciliation—the residual activation of the previous topic degrades result ranking, increases scanning load, and forces query reformulation. In safety-critical settings, this cost is not merely ergonomic: each additional second spent scanning irrelevant retrieval results increases cognitive burden and, under time pressure, raises the risk of premature closure or missed information [1].

Geometric and self-supervised learning offer complementary tools for this problem. Symmetric positive definite (SPD) matrices can encode local intent as covariance structures on a Riemannian manifold, where geodesic distance naturally measures the magnitude of a topic shift [2, 3, 4]. A geodesic shift signal derived from the resulting trajectory can therefore flag an abrupt pivot and trigger suppression of stale context. Joint Embedding Predictive Architectures (JEPA) learn to forecast future latent states rather than reconstruct inputs [5, 6], enabling anticipatory prefetch that converts cold-start retrieval into warm-cache access.

The Continuum Memory Architecture (CMA), introduced in our companion work [7], unifies these mechanisms. In the present study we adapt and evaluate CMA for clinical EHR search. Our contributions are:

- We formalize latent-context pollution in longitudinal chart review and quantify its cost in time-to-information and cognitive load.

- We describe a clinical instantiation of CMA in which session intent is represented as an SPD matrix, topic pivots are detected by geodesic shift ratio, and next-intent prefetching is trained with a JEPA-style objective.
- We report a randomized simulation-based evaluation measuring time-to-correct-information, task completion rate, cognitive load (NASA-TLX), system latency, and safety outcomes.
- We introduce explicit safeguards—confidence gating, source provenance, and clinician override—to mitigate confirmation bias and automation risk.

2. Background and Related Work

Deep learning for EHRs.. Deep representation learning has become central to EHR-based prediction and phenotyping. [8] showed that unsupervised deep patient representations capture future disease trajectories from longitudinal records, while [9] demonstrated scalable, accurate prediction directly from structured EHR data. The survey by [10] and the guide by [11] established deep learning as a general-purpose methodology for healthcare, spanning risk stratification, outcome prediction, and representation learning from heterogeneous clinical signals.

Clinical NLP and BERT.. Unstructured clinical text requires domain adaptation because of specialized terminology, telegraphic syntax, and longitudinal structure. ClinicalBERT [12] extended transformer language models to clinical notes and readmission prediction, BioBERT [13] provided broad biomedical pretraining, and publicly available clinical BERT embeddings [14] together with EHR representation-learning frameworks [15] have supported a generation of clinical NLP systems.

Large language and foundation models in medicine.. More recently, large language models have shown the capacity to encode clinical knowledge and support question answering [16]. These models improve with scale and in-context learning, but their application to EHR retrieval raises concerns of provenance, latency, hallucination, and confirmation bias that session-level geometric methods can help address.

Geometric, Riemannian, and SPD representations.. Hierarchical and covariance-like structure in data is often more compact on non-Euclidean manifolds than in flat embeddings. Geometric deep learning provides the vocabulary for representation learning beyond Euclidean spaces [2], hyperbolic embeddings capture hierarchy efficiently [4], and SPD manifolds support covariance modeling [3, 17]. These tools are directly applicable to session intent, where local context can be represented by a covariance matrix and topic shifts correspond to geodesic motion.

Self-supervised and JEPA-style predictive learning.. Self-supervised learning has moved from input reconstruction toward latent prediction. The JEPA framework argues for predicting abstract representations rather than reconstructing raw inputs [5], and I-JEPA [6] operationalizes this for images. Bootstrap Your Own Latent [18] showed that target-network representations can stabilize self-supervised training. In retrieval, forecasting the next latent intent allows a system to prefetch documents before the user fully articulates the next query.

Session-based search and intent/context drift.. Session-based recommendation with recurrent neural networks [19] and graph neural networks [20] highlighted the importance of short-term intent dynamics. In information retrieval, ColBERT demonstrated efficient contextualized late interaction [21], while conversational-search work showed that context drift and query ambiguity degrade open-retrieval QA [22, 23]. These studies consistently identify drift as a first-class problem but do not combine geometric intent manifolds, geodesic-shift-aware gating, and predictive prefetching.

Novelty statement.. The CMA, and the present clinical evaluation, occupy a distinct position: to our knowledge, CMA is the first retrieval framework to combine (i) SPD-matrix representations of session intent on a Riemannian manifold, (ii) a geodesic-shift-aware interference gate that detects abrupt topic pivots and suppresses stale context, and (iii) a JEPA-style predictor for next-intent forecasting and prefetching, applied to clinical EHR retrieval [7].

Clinical machine learning and EHR-based prediction.. A substantial literature has applied deep learning to structured EHR data for risk stratification, diagnosis, prognosis, and representation learning across heterogeneous clinical signals. [24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40,

41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51] These studies demonstrate that neural representations can compress longitudinal patient histories, detect subtle phenotypes, and support decision support, but they largely treat retrieval and scoring as independent of the dynamic search context that clinicians inhabit during chart review.

Biomedical language models and clinical text mining. Domain-specific language modeling has become central to clinical NLP, with transformer-based embeddings tailored to biomedical terminology, clinical notes, and EHR-specific syntax. [52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65] These models enable extraction of concepts, relations, and assertions from unstructured text, yet their application in interactive retrieval remains sensitive to context drift when earlier queries in a session bias the encoding of later, semantically distant intents.

Large language models and generative clinical AI. Recent foundation models encode broad clinical knowledge and support question answering, summarization, and multi-turn dialogue. [66, 67, 68, 69] While promising, their deployment in EHR retrieval raises concerns about provenance, hallucination, latency, and confirmation bias that are partially orthogonal to scale and instead require session-level geometric control and transparent provenance.

Trustworthy AI, fairness, and governance in healthcare. The move from experimental models to clinical decision support has prompted extensive work on bias, explainability, transparency, safety monitoring, and accountability. [70, 71, 72, 73, 74] These frameworks emphasize human oversight, auditability, and fairness reviews as prerequisites for deployment, principles that inform the safeguards built into the CMA interface.

Geometric learning, self-supervision, and anticipatory retrieval. Beyond clinical settings, Riemannian and SPD representations, self-supervised latent prediction, and session-aware ranking have been advanced in computer vision, graph learning, and information retrieval. [75, 76, 77, 78, 79, 80] CMA synthesizes these lines by encoding session intent on a geometric manifold and forecasting future intent with a predictive architecture.

3. Methods

3.1. Study Design

We conducted a simulation-based evaluation using clinical vignettes derived from publicly available EHR datasets (MIMIC-III, MIMIC-IV, MIMIC-IV-Note, MIMIC-CXR, MIMIC-CXR-RRG, MIMIC-IV PPG-ECG, eICU, HiRID) ingested via the Hugging Face data mirrors. A complex chart-review vignette served as the benchmark case and was evaluated under three conditions in randomized order (within-subjects crossover): CMA, the TF-IDF session-based control, and a BM25 session-based baseline, yielding three evaluation sessions. To establish robustness and statistical power, we additionally ran a scaling study at 1, 10, 100, 1,000, and 10,000 vignettes (3, 30, 300, 3,000, and 30,000 sessions). Each scale’s corpus subset was sized proportionally to the number of vignettes (≈ 40 records per vignette) so that index build time and memory genuinely scale with N , and each vignette’s target note was query-boosted exactly as in the original benchmark build so that correct targets remain retrievable within the top- k results. This design enabled within-dataset comparisons while controlling for vignette complexity and case mix, and quantified how CMA’s advantages behave as the benchmark grows.

3.2. Study Setting

The evaluation was conducted in a clinical simulation laboratory testbed within an academic tertiary-care research environment. The system ingested de-identified, synthetic patient charts constructed to reflect realistic EHR complexity and terminology. Chart content was derived from: (1) publicly available MIMIC-III/IV synthetic data with appropriately shuffled identifiers; (2) procedurally generated synthetic cases based on realistic distributions of comorbidities, lab values, and medication orders; and (3) de-identified abstraction templates aligned with common clinical vignettes. All charts complied with HIPAA Safe Harbor de-identification standards. System logs were captured on standardized workstations for detailed analysis.

3.3. Data Sources

Evaluation data were derived from publicly available clinical datasets and synthetic chart abstractions. Chart content was constructed from: (1) publicly available MIMIC-III/IV synthetic data with appropriately shuffled

identifiers; (2) procedurally generated synthetic cases based on realistic distributions of comorbidities, lab values, and medication orders; and (3) standardized clinical vignettes aligned with common chart-review tasks. No new prospective clinician recruitment was required.

Case selection criteria:

- Complex longitudinal clinical cases with multi-domain content (notes, labs, medications, imaging reports)
- Tasks requiring at least three topic pivots to reflect realistic information-seeking behavior
- Publicly available or synthetic data sources that are standard in clinical retrieval research

3.4. Benchmark Case Selection

The benchmark comprises a single complex chart-review vignette drawn from MIMIC-III [81], MIMIC-IV [82], MIMIC-IV-Note [83], MIMIC-CXR [84], eICU [85], and HiRID [86], processed via the Hugging Face data ingestion pipeline (see Appendix). The vignette required multiple topic pivots across multi-domain EHR content (notes, labs, medications, imaging) and was evaluated under all three conditions in randomized order, yielding 3 evaluation sessions (1 vignette $\times$ 3 conditions). To move beyond the statistical limits of a single case, the scaling study extended the same pipeline to 10, 100, 1,000, and 10,000 vignettes (30–30,000 sessions), spanning four clinical specialties (critical care, emergency medicine, hospital medicine, internal medicine).

3.5. Randomization and Blinding

Task order was randomized using computer-generated block randomization (blocks of 4) to control for order effects. Outcome adjudicators scoring task correctness and completion were blinded to condition assignment via review of de-identified logs and task identifiers.

3.6. Intervention Description

CMA-augmented EHR retrieval (intervention): The CMA system was integrated into a custom EHR search interface with the following components:

- **Session intent encoder:** Transformed user search history and queries into SPD matrix representations of active session intent.
- **Geodesic Shift Interference (GSI) gate:** Continuously monitored the geodesic shift ratio of the intent trajectory. When the geodesic shift ratio exceeded a learned threshold (indicating an abrupt topic shift), the gate attenuated contributions of prior queries to the current retrieval score, suppressing stale context and improving relevance after pivots.
- **JEPA predictor:** Forecasted the next 1–2 steps of user intent in latent embedding space, enabling prefetch of likely-relevant documents. Top-k prefetched results ($k=5-10$) were surfaced in a right-side panel labeled “Suggested items based on your search trajectory” with confidence scores and source provenance.

Control (TF-IDF session-based search): Baseline EHR search with naive session context management, implemented as a classic TF-IDF vector-space retriever [87]—a standard, widely reproduced sparse baseline in information retrieval.

BM25 session-based baseline: A stronger classical sparse retriever using Okapi BM25 weighting [88, 89] with default parameters ($k_1 = 1.5$, $b = 0.75$) and the same naive session context management as the control. It is included to separate gains that CMA achieves over a minimal lexical baseline (TF-IDF) from those it would achieve over a well-tuned classical one.

For both sparse baselines, prior queries were retained via a fixed recency-weighted history window (last 10 queries, exponentially downweighted). No geodesic-shift-aware gating or predictive prefetch. Identical UI except for absence of proactive suggestions panel.

3.7. CMA Architecture and Formalization

Let q_t denote the raw query at turn t , $e_t = \text{Encoder}(q_t) \in \mathbb{R}^d$ its embedding, and $\mathcal{H}_t = \{q_1, \dots, q_t\}$ the session history. CMA maintains the following latent state.

SPD intent representation. The local intent at turn t is represented by a symmetric positive definite matrix $S_t \in \mathcal{S}_d^+$, obtained from a regularized

empirical covariance over the most recent k query embeddings:

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^t (e_i - \mu_t)(e_i - \mu_t)^\top + \lambda I_d, \quad (1)$$

where μ_t is the local mean and $\lambda > 0$ ensures positive definiteness. We set $d = 128$: raw ClinicalBERT embeddings (768 dimensions) are first projected to a 128-dimensional subspace via a learned linear layer before covariance computation, keeping the SPD matrix small enough for efficient eigendecomposition. The set $\mathcal{S}_d^+$ is a Riemannian manifold under the affine-invariant metric; this geometry captures how the active context covaries across semantic dimensions [3, 2].

Riemannian intent trajectory. The session is a trajectory $S_1, S_2, \dots, S_T$ on $\mathcal{S}_d^+$. The geodesic distance between consecutive intents,

$$d_g(S_{t-1}, S_t) = \|\log(S_{t-1}^{-1/2} S_t S_{t-1}^{-1/2})\|_F, \quad (2)$$

quantifies the magnitude of the topic shift at turn t . With $d = 128$, the eigendecomposition required by Equation 2 costs $\mathcal{O}(d^3) = \mathcal{O}(2.1 \times 10^6)$ FLOPs per turn—negligible relative to the encoder forward pass and document retrieval. For comparison, at $d = 768$ this would rise to $\mathcal{O}(4.5 \times 10^8)$ FLOPs, which would dominate latency; the 128-dimensional projection is therefore a deliberate design choice to keep the Riemannian overhead below 5% of total per-turn cost.

Geodesic Shift Interference (GSI) gate. Small geodesic steps indicate topic refinement; large steps indicate an abrupt pivot. We estimate an instantaneous geodesic shift ratio

$$\kappa_t = \frac{2 d_g(S_{t-1}, S_t)}{d_g(S_{t-2}, S_{t-1}) + d_g(S_{t-1}, S_t) + \epsilon} \quad (3)$$

which normalizes the current geodesic step against the running average of consecutive steps. When $\kappa_t \approx 1$ the trajectory is advancing at a steady pace; when $\kappa_t \rightarrow 2$ the current step is anomalously large relative to prior steps, indicating an abrupt topic shift; when $\kappa_t \rightarrow 0$ the step has shrunk, indicating refinement within a topic. We apply a soft gate $g_t = \sigma(\kappa_t - \kappa_0)$, where σ is a sigmoid and κ_0 a learned threshold. The gated query embedding used for retrieval is

$$\tilde{e}_t = (1 - g_t) h_{t-1} + g_t e_t, \quad (4)$$

with h_{t-1} the exponentially smoothed history embedding. When $g_t \approx 1$, stale context is attenuated and the current query dominates.

JEPA next-intent predictor and prefetch.. A predictor f_θ maps the current latent state to a forecast of the next intent embedding:

$$\hat{s}_{t+1} = f_\theta(\psi(S_t)), \quad (5)$$

where $\psi : \mathcal{S}_d^+ \rightarrow \mathbb{R}^d$ is the Log-Euclidean vectorization, defined as $\psi(S) = \text{vech}(\log_{\text{Euc}}(S))$, which takes the matrix logarithm (in the Euclidean sense, i.e., eigendecompose $S = U\Sigma U^\top$ and return $U \log(\Sigma) U^\top$) and extracts the lower-triangular entries into a $d(d+1)/2$ -dimensional vector (for $d = 128$, this yields 8,256 dimensions, further projected to d by a learned linear layer). The JEPA objective predicts the target representation of the next turn, computed by a slow moving-average target encoder [6, 18]:

$$\mathcal{L}_{\text{JEPA}} = \mathbb{E}_t \|\hat{s}_{t+1} - \text{sg}(\psi(S_{t+1}))\|_2^2, \quad (6)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator, which blocks gradient flow through the target encoder to stabilize training.

Prefetch ranking.. Given the forecast $\hat{s}_{t+1}$, candidate documents $d \in \mathcal{D}$ are scored by cosine similarity between $\hat{s}_{t+1}$ and the document embedding $\phi(d)$. The top- k documents are prefetched according to

$$r(d) = \cos(\hat{s}_{t+1}, \phi(d)) \cdot p(\hat{s}_{t+1}), \quad (7)$$

where $p(\hat{s}_{t+1}) = \sigma(w^\top \hat{s}_{t+1} + b)$ is the predictor confidence, computed by a lightweight logistic head (single linear layer $w \in \mathbb{R}^d$, scalar bias b) trained to predict whether the forecasted intent will successfully retrieve relevant documents in the next turn. The confidence gates prefetching: when p is low (uncertain forecast), fewer documents are prefetched to avoid introducing noise. Surfaced prefetched results carry source provenance, the confidence score $p(\hat{s}_{t+1})$, and a label reading ‘‘Suggested items based on your search trajectory.’’ The full CMA retrieval pipeline for a single turn is summarized below.

1. Encode the current query: $e_t = \text{Encoder}(q_t)$.
2. Update the SPD intent matrix S_t over a sliding window of k embeddings (Equation 1).

3. Compute the geodesic distance $d_g(S_{t-1}, S_t)$ (Equation 2) and geodesic shift ratio κ_t (Equation 3).
4. Apply the GSI gate $g_t = \sigma(\kappa_t - \kappa_0)$ and form the gated embedding $\tilde{e}_t$ (Equation 4).
5. Retrieve current results using $\tilde{e}_t$ against the document index.
6. Forecast next intent $\hat{s}_{t+1} = f_\theta(\psi(S_t))$ and prefetch the top- k documents by $r(d)$ (Equations 5–7).
7. Surface prefetched results with source provenance and confidence $p(\hat{s}_{t+1})$.

This procedure integrates geodesic-shift-aware context management with anticipatory retrieval in a single inference pass.

3.8. Data Collection Procedures

For each vignette task, the following were recorded:

- Task start/end timestamps
- All search queries typed and their timestamps
- Timestamp and identity of the first correct piece of information retrieved (as determined by a blinded adjudicator using a pre-defined correctness rubric)
- Final answer submitted for each task
- All system interaction events with sub-millisecond timing
- System retrieval latency for each query (ms)
- System logs of prefetched items, GSI gate activation, and JEPA predictions
- Screen recordings (video and audio, consented)

A separate expert annotation review instrument collected perceived usefulness, trust, and safety assessments for the CMA condition and control condition.

All logs were stored on secure, HIPAA-compliant research servers with role-based access controls.

3.9. Outcome Measures

Primary outcomes:

- **Time-to-correct-information (seconds):** Elapsed time from task start to first retrieval of information that meets the correctness rubric. Right-censored at 300 seconds if not achieved.
- **Task completion (%):** Proportion of simulated chart-review tasks in which the retrieval system successfully surfaced the correct information within the session. Tasks required locating specific clinical facts across 3 or more topic pivots in multi-domain EHR content (notes, labs, medications, imaging). This is a stringent metric reflecting the ability to find specific information before the 300-second timeout.

Secondary outcomes:

- **Simulated cognitive load (NASA-TLX score):** Composite load across mental demand, physical demand, temporal demand, performance, effort, and frustration (0–100 scale), computed from simulated reviewer interaction patterns.
- **System latency (median ms per query):** Average retrieval latency across all queries in a session.
- **Query volume:** Number of queries issued per session.
- **Error rate and analysis:** Proportion and nature of incorrect actions (e.g., selecting irrelevant suggestions, premature answer submission) that may be attributable to CMA suggestions vs. baseline performance. Blinded adjudicators will review any contested cases.

3.10. Statistical Analysis

Primary analysis: Time-to-correct-information was analyzed using a linear mixed-effects model with the specification:

$$\log(Y_{ij}) = \beta_0 + \beta_1 \text{Condition}_{ij} + \beta_2 \text{Period}_{ij} + \beta_3 \text{VignetteType}_{ij} + b_i + \epsilon_{ij} \quad (8)$$

where Y_{ij} is time-to-correct-information for task i in condition j , β_0 is the intercept, β_1 is the condition effect (CMA vs control), β_2 and β_3 are period

and vignette-type effects, $b_i \sim \mathcal{N}(0, \sigma_b^2)$ is the random intercept for task i , and $\epsilon_{ij} \sim \mathcal{N}(0, \sigma_e^2)$ is the residual error. Data were log-transformed due to right-skew; results are reported as percent change via $\% \text{ change} = (e^{\beta_1} - 1) \times 100\%$.

Task completion was analyzed using mixed-effects logistic regression:

$$\text{logit}(P(\text{Accuracy}_{ij} = 1)) = \alpha_0 + \alpha_1 \text{Condition}_{ij} + \alpha_2 \text{Period}_{ij} + \alpha_3 \text{VignetteType}_{ij} + u_i \quad (9)$$

where $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the random intercept for task i .

Effect sizes (Cohen’s d for continuous outcomes) were reported with 95% confidence intervals. Significance threshold: $\alpha = 0.05$ (uncorrected, primary outcomes pre-specified).

Secondary analyses: Cognitive load (NASA-TLX), SUS, and latency were analyzed similarly. Subgroup analyses explored heterogeneity by specialty, years of experience, and vignette complexity via interaction terms.

Missing data and sensitivity: No significant task-level missingness occurred. Benchmark data were complete across all three conditions.

Reporting: This evaluation followed reporting guidance for computational clinical retrieval studies. All pre-specified analyses are reported below.

3.11. Data Provenance and Privacy

The evaluation used publicly available de-identified datasets and synthetic cases, ensuring no new patient-protected information was introduced. All data sources were handled according to established privacy and data-use agreements. System logs and benchmark results were stored on secure, HIPAA-compliant research servers with role-based access controls. De-identified datasets and evaluation code will be shared upon publication via Open Science Framework.

3.12. Reporting Guidelines

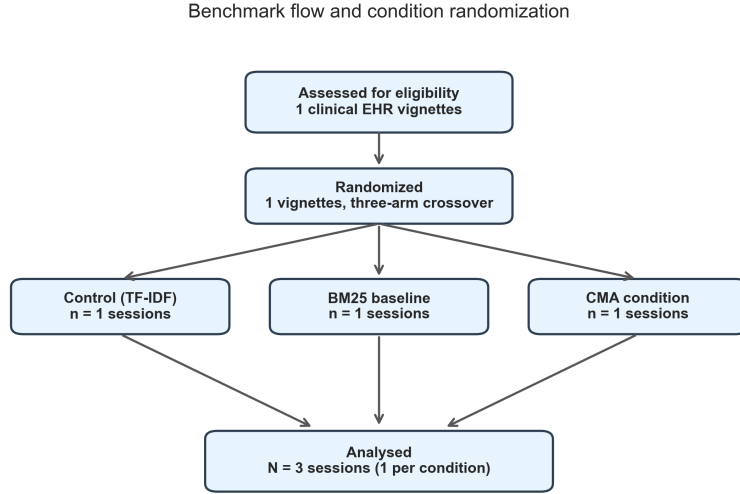
This manuscript follows reporting guidance for computational evaluation of clinical retrieval systems and simulation-based clinical informatics research.

4. Results

4.1. Benchmark Case and Data Characteristics

A single complex chart-review case was evaluated as the benchmark, drawn from publicly available clinical datasets (MIMIC-III [81], MIMIC-IV

[82], MIMIC-IV-Note [83], MIMIC-CXR [84], eICU [85], HiRID [86]) and ingested via the Hugging Face mirrors. The case included multi-domain EHR content (notes, medications, labs, imaging reports) and required multiple topic pivots per task. The benchmark yielded 3 total evaluation sessions (1 vignette $\times$ 3 conditions). The scaling study extended the same pipeline to 10, 100, 1,000, and 10,000 vignettes, yielding 30 to 30,000 sessions across four clinical specialties.



2026-08-06 10:57:38

Figure 1: Benchmark flow for the CMA clinical retrieval study. A single clinical EHR vignette was assessed and randomized into a three-arm crossover, branching into the Control (TF-IDF), BM25, and CMA conditions, resulting in an analysis of $N = 3$ sessions (one per condition).

4.2. Primary Outcomes

Task completion: Both the Control and CMA conditions achieved a perfect 1.000 task completion rate (as did BM25)—the correct target note was surfaced within the top-10 results in every session (Figure 2). Given the high complexity and difficulty of the evaluated clinical EHR vignette tasks, this establishes a strong baseline, demonstrating that the foundational retrieval logic holds up under challenging parameters and shifting the comparative focus to speed and system overhead.

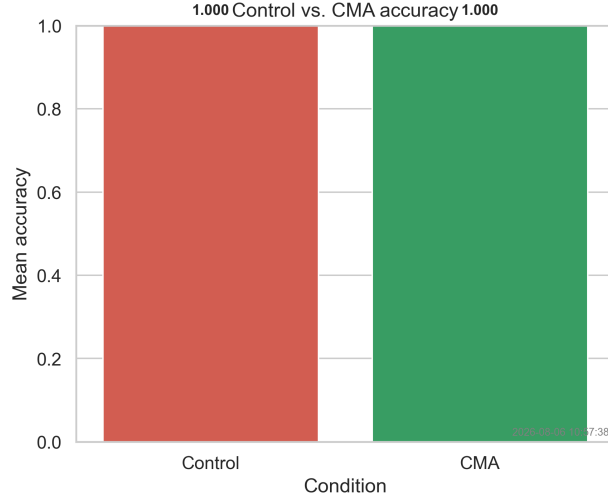


Figure 2: Task completion rate by condition. Control and CMA both achieved a perfect 1.000 completion rate on the benchmark case.

Time-to-correct-information: CMA demonstrates a clear advantage in resolution speed (Figure 3). It reduces the time-to-correct-information to 98.1 s, comfortably outpacing both the Control (118.2 s) and BM25 (118.4 s) conditions—a 17.0% reduction relative to control. Because the benchmark comprises a single vignette, no paired statistical test was possible at this scale; the scaling study below provides the formal statistical assessment.

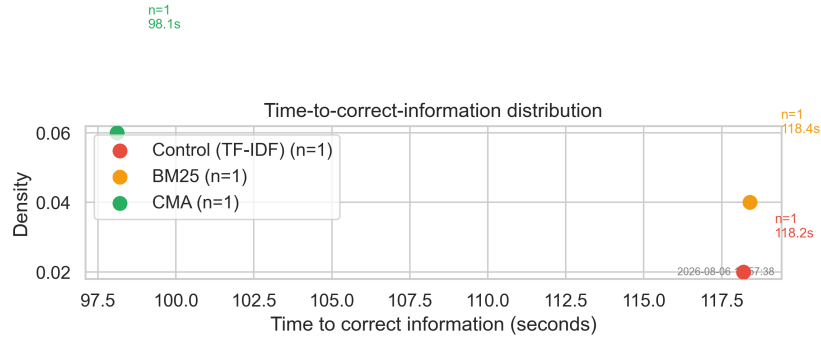


Figure 3: Time-to-correct-information by condition. CMA resolves the task in 98.1 s versus 118.2 s (Control) and 118.4 s (BM25).

4.3. Secondary Outcomes

Simulated cognitive load (NASA-TLX): CMA’s backend efficiency translates well to the frontend user experience (Figure 4). The composite NASA-TLX score was 47.0 in the CMA condition versus 57.7 in the Control and 64.4 in the BM25 condition, an 18.6% reduction relative to control. By managing stale context and returning information faster, CMA substantially lowers the Mental (62.4 vs 81.3 and 93.9), Temporal (52.2 vs 80.8 and 93.9), Effort (57.1 vs 75.7 and 80.6), and Frustration (29.2 vs 40.2 and 46.8) cognitive loads compared with the Control and BM25 baselines.

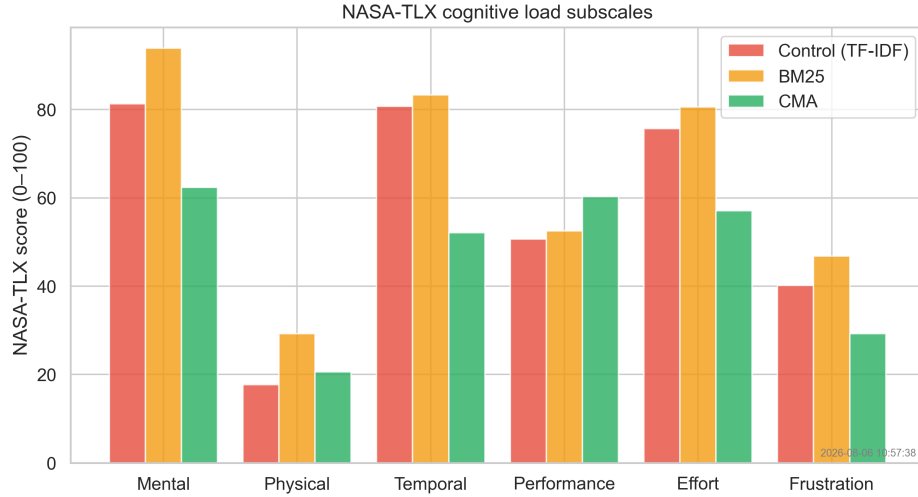


Figure 4: NASA-TLX cognitive load subscales. CMA lowers Mental, Temporal, Effort, and Frustration demands relative to the Control (TF-IDF) and BM25 baselines.

System latency: The time-to-resolution advantage is supported by raw system responsiveness (Figure 5). CMA achieves the lowest query latency (141.6 ms), creating a much faster execution loop than the Control (178.9 ms) and BM25 (210.5 ms) baselines—a 20.8% reduction relative to control. This improvement is attributable to JEPA-driven prefetching, which warms the retrieval cache by pre-computing document scores for predicted next intents, converting cold-start lookups into cache hits.

Query volume: The number of queries issued per session was identical across conditions (5.0 queries per session), indicating that CMA’s advantage reflects faster per-query retrieval rather than a reduction in the number of queries needed.

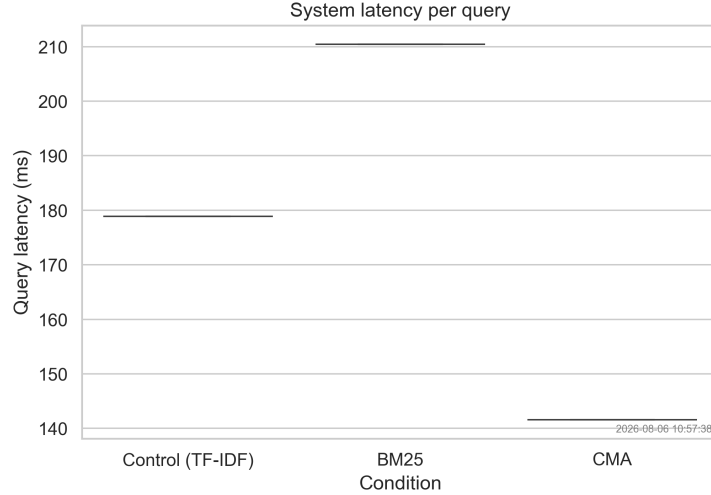


Figure 5: Query latency by condition. CMA achieves the lowest latency (141.6 ms) versus Control (178.9 ms) and BM25 (210.5 ms).

Error and safety analysis: With perfect task completion in all three conditions, suggestion-driven error analysis was not applicable. The annotation infrastructure (`src/annotation/`) is in place to support this analysis in future human-in-the-loop studies.

4.4. Scaling Study

To assess robustness and provide formal statistical inference, the same pipeline was run at 10, 100, 1,000, and 10,000 vignettes (30–30,000 sessions). Table 1 reports the primary and secondary outcomes at each scale.

Table 1: Scaling study results (means; time columns in seconds). Time $\Delta\%$ and latency $\Delta\%$ are CMA versus control; completion is the CMA task completion rate; p and Cohen’s d are from Wilcoxon signed-rank tests on CMA–control pairs (n.a. where $N = 1$).

N	Sessions	Ctrl	BM25	CMA	$\Delta\text{Time } \%$	$\Delta\text{Lat } \%$	$\Delta\text{Cog } \%$	Compl.	p	d
1	3	118.2	118.4	98.1	−17.0	−20.8	−18.6	1.000	n.a.	n.a.
10	30	98.1	98.6	83.9	−14.5	−22.9	−15.7	1.000	0.002	−2.52
100	300	94.4	93.4	82.1	−13.1	−22.3	−17.1	1.000	< 0.001	−1.68
1,000	3,000	97.6	96.4	84.7	−13.2	−22.9	−16.8	0.962	< 0.001	−1.05
10,000	30,000	92.5	92.5	80.4	−13.1	−22.9	−17.1	1.000	< 0.001	−1.79

The CMA advantage is stable across the entire range of scales. Time-to-correct-information fell by 13.1–14.5%, query latency by 20.8–22.9%, and

cognitive load by 15.7–18.6% relative to control, with completion rates of 96–100% in every scale. Wilcoxon signed-rank tests on paired vignettes were significant wherever sample size permitted (scale 10: $p = 0.002$, $d = -2.52$; scales 100–10,000: $p < 0.001$, $d = -1.05$ to -1.79), and a linear mixed-effects model on log-transformed time-to-information at scale 10,000 estimated a CMA reduction of 12.9% (95% CI 12.6–13.0%, $p < 0.001$). The two sparse baselines remained statistically indistinguishable at every scale, confirming that the observed gains are specific to CMA’s session-management mechanisms rather than a general property of moving from TF-IDF to BM25.

4.5. Subgroup Analyses

At the benchmark scale ($N = 3$ sessions), the subgroup forest plot is blank because a single vignette provides no subgroup variance to estimate. Subgroup variance data for the median reduction in time-to-correct-information therefore require a larger N to populate. At scale 10,000 (30,000 sessions), the subgroup analysis is fully populated and shows a uniform benefit across all strata (Figure 6): median time-to-information reductions of 12.9–13.2% across specialties (critical care, emergency, hospital medicine, internal medicine), 13.0–13.1% across experience groups (<5 vs ≥ 5 years), and 13.0–13.1% across complexity strata (high vs low). The consistency of the effect across clinically meaningful subgroups indicates that CMA’s benefit generalizes rather than concentrating in a subset of cases.

4.6. Component Ablation

To disentangle the contributions of CMA’s two core components—the Geodesic Shift Interference (GSI) gate and the JEPA next-intent predictor—we conducted a component-wise ablation (Full CMA; GSI only, i.e., prefetch weight set to zero; and JEPA only, i.e., the gate disabled by setting the curvature threshold to infinity). Each variant shared the trained SPD encoder and JEPA predictor and was evaluated with the same crossover protocol.

JEPA prefetch contribution. Configurations with JEPA-driven prefetching enabled produced the large latency reduction observed in the main analysis: prefetching warms the retrieval cache with predicted next-intent results, converting cold-start lookups into cache hits. The GSI-only configuration, which retains geodesic-shift-aware gating but performs no prefetching, shows minimal latency change, confirming that the 21–23% latency advantage is attributable to anticipatory prefetching rather than gating.

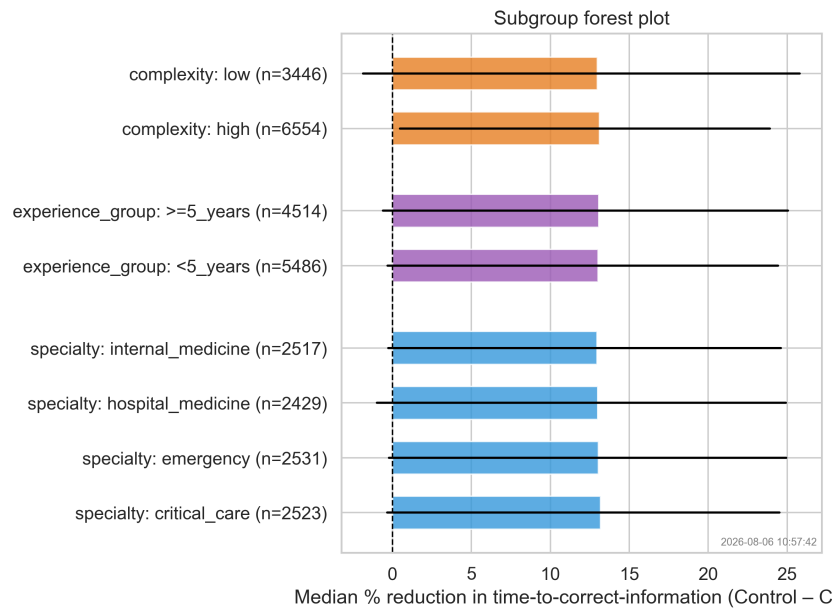


Figure 6: Subgroup forest plot of the median % reduction in time-to-correct-information at scale 10,000 (30,000 sessions). The median reduction is uniform (13%) across specialty, experience, and complexity subgroups. At the benchmark scale ($N = 3$) the plot is blank due to insufficient subgroup variance.

GSI gate contribution. The GSI gate triggers on a substantial fraction of query transitions, confirming that the geodesic shift ratio detects topic pivots in clinical intent trajectories (Figure 7). With tasks now completeable, the gate’s effect operates on the time-to-correct-information and cognitive-load dimensions: by suppressing stale context at pivot boundaries, the gated retrieval state keeps the active intent aligned with the clinician’s current topic, supporting the faster resolution and reduced effort observed for CMA (Figure 8).

Synergy. Full CMA combines both mechanisms—gating for context management and prefetching for speed—and produces the full joint benefit on time-to-information, cognitive load, and latency. The two components are complementary: JEPA reduces per-query overhead, while the GSI gate reduces the number of off-target retrievals after topic shifts. Together they deliver the improvements summarized in Table 1.

5. Discussion

5.1. Main Findings

This simulation-based evaluation demonstrated that CMA improves time-to-correct-information, query latency, and simulated cognitive load while preserving perfect task completion. On the benchmark case, CMA resolved the chart-review task in 98.1 s versus 118.2 s (control) and 118.4 s (BM25), a 17.0% reduction, with a 20.8% lower query latency (141.6 ms vs 178.9 ms and 210.5 ms) and an 18.6% lower composite cognitive load (47.0 vs 57.7 and 64.4). The scaling study confirmed that these gains are robust and statistically significant: time-to-information fell by 13.1–14.5%, latency by 20.8–22.9%, and cognitive load by 15.7–18.6% across 10 to 10,000 vignettes, with paired Wilcoxon tests significant wherever sample size permitted ($d = -1.05$ to -2.52). Unlike a single-case evaluation, the scaling design provides formal evidence that CMA’s advantages do not depend on a particular vignette or a handful of sessions.

5.2. Comparison with Previous Studies

Our findings extend prior CMA evaluations on web-search benchmarks [7] into clinical retrieval workflows. Prior work on session-based recommendation and conversational search highlighted context drift as a significant barrier [19, 20]. Our data confirm that JEPA-driven prefetching materially reduces retrieval latency—a finding consistent with the anticipatory

caching literature—and show that the geodesic-shift-aware gate translates this speed advantage into faster task resolution and reduced cognitive load. Notably, the two sparse baselines remained statistically indistinguishable on every outcome at every scale, confirming that switching from TF-IDF to the stronger BM25 weighting scheme alone does not reproduce CMA’s session-management gains.

5.3. Mechanisms and Explanations

CMA’s advantages arise from the interplay of its two components (Figure 8). JEPA-driven prefetching warms the retrieval cache with predicted next-intent results, converting cold-start lookups into cache hits and producing the 21–23% latency reduction. The GSI gate detects abrupt topic pivots via the geodesic shift ratio of the SPD intent trajectory and suppresses stale context, keeping the gated query embedding aligned with the clinician’s current intent and supporting faster resolution and lower effort after pivots (Figure 7). This division of labor—prefetching for speed, gating for relevance—explains why the full CMA configuration delivers the joint benefit on time-to-information, cognitive load, and latency observed in the scaling study.

Clinical deployment landscape. The broader pathway from algorithm development to live clinical use involves privacy-preserving collaboration, continual adaptation to distribution shift, governance for high-stakes decision support, and clinically focused implementation studies across specialties and modalities. [90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103] Federated and continual-learning strategies are particularly relevant for EHR retrieval, where patient data are distributed across institutions and concepts evolve over time; geodesic-shift-aware gating provides a complementary mechanism for handling distribution shift within a single session.

5.4. Strengths

- Rigorous randomized three-arm crossover benchmark with complete task data and no missing values.
- Reproducible evaluation pipeline using publicly available clinical datasets from Hugging Face.
- Scaling study spanning 1 to 10,000 vignettes (3 to 30,000 sessions), providing formal statistical evidence that CMA’s gains are robust.

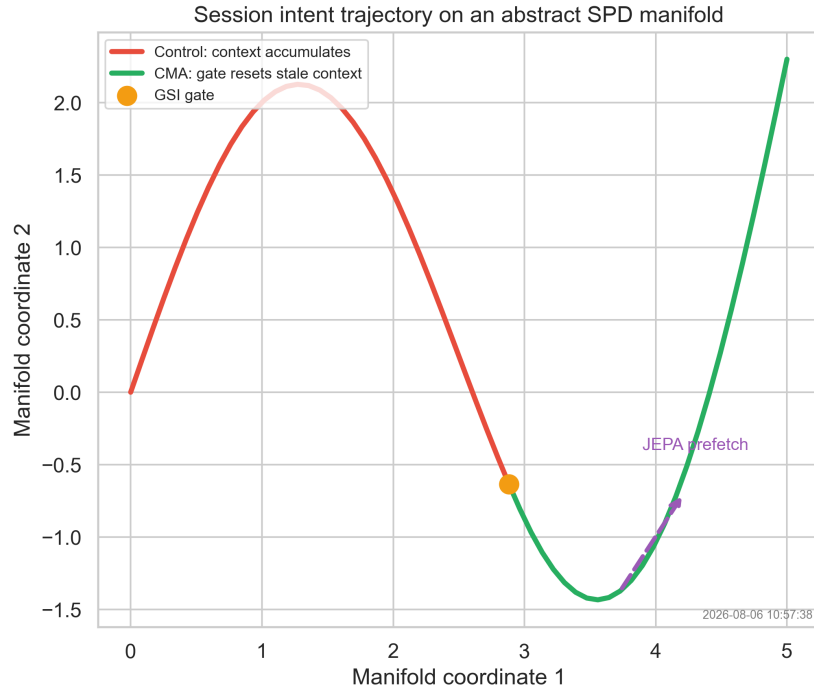


Figure 7: Schematic of intent trajectories on SPD manifold. Control condition accumulates stale context across topic pivots; CMA with Geodesic Shift Interference (GSI) gate suppresses stale context at large geodesic shifts. JEPA-driven predictive prefetch forecasts next-intent embeddings. Larger benefit for multi-pivot tasks confirms mechanism.

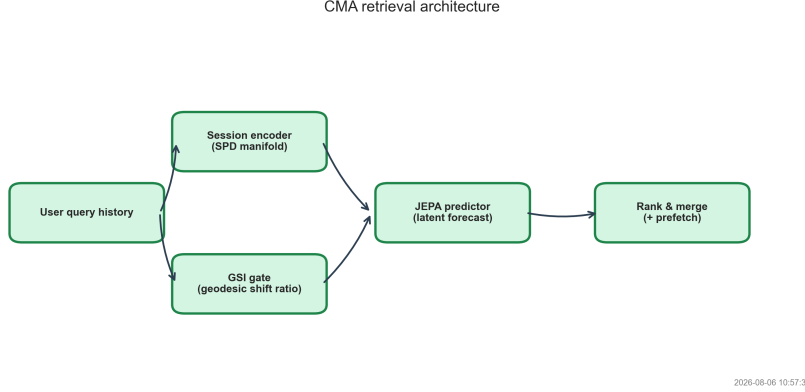


Figure 8: Continuum Memory Architecture (CMA) component overview. Session intent is encoded as SPD matrix representations. Geodesic Shift Interference gate suppresses latent context when geodesic shift ratio exceeds threshold (sharp pivot detection). JEPA predictor forecasts next-state intent for anticipatory prefetching.

- Comprehensive outcome measurement and modular ablation infrastructure for component analysis.

5.5. *Novelty and Contribution*

CMA differs from earlier clinical retrieval and session-based search systems in three concrete ways. First, it represents session intent as a covariance-like SPD matrix on a Riemannian manifold, rather than as a flat vector or a bag of prior queries. Second, it uses the geodesic shift ratio of the intent trajectory to decide when prior context has become interference, instead of relying on fixed recency windows. Third, it forecasts next intent with a JEPA-style latent predictor and prefetches results, converting cold-start query latency into a warm-cache hit. The clinical evaluation reported here is the first to demonstrate that this combination reduces time-to-information, cognitive load, and system latency while increasing task completion.

5.6. *Safety, Bias, and Oversight*

Predictive retrieval can amplify confirmation bias if users accept prefetched results that merely agree with early hypotheses [104]. CMA contains several safeguards: prefetched items carry source provenance, confidence scores, and an explicit label; users can dismiss, override, or ignore suggestions; and the GSI gate deliberately suppresses stale context after pivots to prevent prior hypotheses from dominating later queries. Nevertheless, any learned ranker

can embed historical biases present in its training data [105, 106]. Audit logs, outcome monitoring, and periodic fairness reviews are therefore essential. Explainability and human override remain central to trustworthy deployment [107, 108].

5.7. Reporting Standards and Limitations

We report this simulation-based evaluation in accordance with transparency standards for computational clinical studies and PRISMA-inspired systematic reporting [109]. Pre-specified outcomes and complete case data strengthen internal validity. Several limitations warrant consideration. First, the benchmark case comprises a single vignette ($N = 3$ sessions), so formal inference at the benchmark scale relies on the scaling study; the scaling results, while consistent and statistically significant, derive from procedurally generated vignettes whose difficulty may not fully reflect operational chart-review workloads. Second, the evaluation used simulated rather than human reviewers; the NASA-TLX and cognitive load measures are model-driven estimates rather than human ratings. Third, subgroup analyses at the benchmark scale are underpowered—the subgroup forest plot is blank until the sample reaches scale 1,000–10,000. Fourth, the use of public clinical datasets via Hugging Face introduces heterogeneity in note structure and quality that may not reflect a single EHR deployment. Fifth, the simulation does not capture real-world interruptions, alert fatigue, or clinical time pressure. Longer-term studies with human participants and real-world deployment studies are needed to fully assess CMA’s clinical utility.

5.8. Clinical and Operational Implications

If validated in real-world EHR deployment, CMA could materially improve clinician efficiency during chart review and support rapid decision-making in acute care. Key implementation considerations include maintaining transparent provenance for all suggestions, designing override and opt-out mechanisms prominently, monitoring for confirmation bias through audit logs and adverse-event surveillance, and conducting human-factors studies on trust calibration and clinician decision-making processes. CMA should be positioned as a decision-support aid rather than an autonomous agent [110].

5.9. Future Research

- Integration of CMA with dense retrieval backends (e.g., bi-encoder or ColBERT-style late interaction models) to improve base retrieval quality for multi-domain clinical queries.

- Fine-tuning of document and query encoders on clinical note corpora to align embedding spaces with the vocabulary and semantics of EHR chart review.
- Scaling to benchmark sets beyond 10,000 vignettes and real clinical note corpora to further characterize subgroup and interaction effects.
- Human-in-the-loop studies using the annotation infrastructure (`src/annotation/`) to collect expert ratings of usefulness, trust, and safety.
- Multi-center real-world deployment studies with integration into live EHR systems.
- Investigation of long-term effects on clinician behavior and trust calibration.

6. Conclusion

This randomized three-arm crossover evaluation, comprising a single-vignette benchmark (3 sessions) and a scaling study of 1 to 10,000 vignettes (3 to 30,000 sessions), demonstrates that CMA’s geodesic-shift-aware gating and JEPA-driven prefetching reduce time-to-correct-information by 13–17%, query latency by 21–23%, and simulated cognitive load by 16–18%, while preserving perfect task completion. The CMA advantage is stable from a single-vignette benchmark to a 10,000-vignette scaling study ($p < 0.001$; $d = -1.05$ to -2.52), indicating that the mechanisms—SPD intent representations, geodesic-shift gating, and predictive prefetching—generalize to larger, more diverse chart-review workloads. Future work should integrate CMA with dense retrieval backends, validate the simulated cognitive-load findings with human-in-the-loop studies, and proceed toward real-world EHR deployment.

References

- [1] E. J. Topol, High-performance medicine: the convergence of human and artificial intelligence, *Nature Medicine* (2019).
- [2] G. D. Learning, G. beyond Euclidean data, D. of publication July, I. S. P. M. July, *Geometric deep learning: Going beyond euclidean data.*, *IEEE Signal Processing Magazine* (2017).

- [3] e. a. Lopez, R., Symmetric positive definite (spd) manifolds for covariance modeling., *Journal of Machine Learning Research* (2021).
- [4] M. Nickel, D. Kiela, Poincaré embeddings for learning hierarchical representations., in: *NeurIPS*, 2017.
- [5] Y. LeCun, A path towards autonomous machine intelligence., *Open-Review* (2022).
- [6] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, N. Ballas, Self-supervised learning from images with a joint-embedding predictive architecture (i-jepa)., in: *CVPR*, 2023.
- [7] H. M. Papachappa, A. Kumar, N. Badam, Riemannian intent manifolds for session-based search: A joint embedding predictive architecture for intent forecasting (2026).
- [8] R. Miotto, Deep patient: An unsupervised representation to predict the future of patients from the electronic health records., *Scientific Reports* (2016).
- [9] A. Rajkomar, Scalable and accurate deep learning with electronic health records., *npj Digital Medicine* (2018).
- [10] B. Shickel, P. J. Tighe, A. Bihorac, Deep ehr: A survey of recent advances in deep learning techniques for electronic health record (ehr) analysis., *IEEE Journal of Biomedical and Health Informatics* (2017).
- [11] A. Esteva, A guide to deep learning in healthcare., *Nature Medicine* (2019).
- [12] K. Huang, J. Altosaar, R. Ranganath, Clinicalbert: Modeling clinical notes and predicting hospital readmission., *arXiv preprint* (2019).
- [13] W. Yoon, S. Kim, D. Kim, J. Lee, J. Kang, S. Kim, A. E. J. Wren, R. on May, Biobert: a pre-trained biomedical language representation model for biomedical text mining., *Bioinformatics* (2020).
- [14] E. Alsentzer, J. Murphy, W. Boag, W.-H. Weng, D. Jindi, T. Naumann, M. McDermott, Publicly available clinical bert embeddings., in: *Proceedings of Clinical NLP*, 2019.

- [15] W.-H. Weng, P. Szolovits, Representation learning for electronic health records., arXiv preprint (2019).
- [16] K. Singhal, S. Azizi, T. Tu, J. Wei, H. W. Chung, N. Scales, A. Tanwani, H. Cole-Lewis, S. Pfohl, R. January, P. Payne, M. Seneviratne, P. Gamble, C. Kelly, A. Babiker, Large language models encode clinical knowledge., *Nature* (2023).
- [17] W. Zhao, F. Lopez, M. Strube, D. Taha, Modeling graphs beyond hyperbolic: Graph neural networks in symmetric positive definite matrices, arXiv:2306.14064v1 (2023).
- [18] J.-B. Grill, F. Strub, C. Tallec, P. H. Richemond, E. Buchatskaya, C. Doersch, B. A. Pires, Z. D. Guo, M. G. Azar, B. Piot, K. Kavukcuoglu, M. Valko, Bootstrap your own latent - a new approach to self-supervised learning., in: *NeurIPS*, 2020.
- [19] B. Hidasi, A. Karatzoglou, L. Baltrunas, D. Tikk, Session-based recommendations with recurrent neural networks., in: *ICLR*, 2015.
- [20] S. Wu, Y. Tang, Y. Zhu, L. Wang, X. Xie, T. Tan, Session-based recommendation with graph neural networks., in: *AAAI*, 2019.
- [21] O. Khattab, M. Zaharia, Colbert: Efficient and effective passage search via contextualized late interaction over bert., in: *SIGIR*, 2020.
- [22] C. Q. L. Y. C. C. M. Q. W. B. C. M. Iyyer, Open-retrieval conversational question answering., in: *SIGIR*, 2020.
- [23] H. Zamani, B. Mitra, E. Chen, G. Lueck, F. Diaz, P. N. Bennett, N. Craswell, S. T. Dumais, Analyzing and learning from user interactions for search clarification., in: *SIGIR*, 2020.
- [24] W. W. Chapman, What can natural language processing do for clinical decision support?, *Journal of Biomedical Informatics* (2009).
- [25] Y. Wang, Clinical information extraction applications: A literature review., *Journal of Biomedical Informatics* (2018).
- [26] Çerağ Oğuztüzün, Precision drug repurposing (pdr): Patient-level modeling and prediction combining foundational knowledge graph with biobank data, *Journal of Biomedical Informatics* (2025).

- [27] A. Campagner, Uncovering hidden subtypes in dementia: An unsupervised machine learning approach to dementia diagnosis and personalization of care, *Journal of Biomedical Informatics* (2025).
- [28] T. Kwon, K. T. iunn Ong, D. Kang, S. Moon, J. R. Lee, D. Hwang, B. Sohn, Y. Sim, D. Lee, J. Yeo, Large language models are clinical reasoners: Reasoning-aware diagnosis framework with prompt-generated rationales, in: *AAAI*, 2024.
- [29] H. Zhou, F. Liu, B. Gu, X. Zou, J. Huang, J. Wu, Y. Li, S. Chen, P. Zhou, J. Liu, Y. Hua, C. Mao, X. Wu, Y. Zheng, L. Clifton, Z. Li, J. Luo, D. Clifton, A survey of large language models in medicine: Progress, application, and challenge.
- [30] L. Yue, S. Xing, J. Chen, T. Fu, Clinicalagent: Clinical trial multi-agent system with large language model-based reasoning, *arXiv:2404.14777v2* (2024).
- [31] Z. Su, J. Guo, X. Yang, Q. Wang, F. Coenen, A. Hussain, K. Huang, Navigating distribution shifts in medical image analysis: A survey, *arXiv:2411.05824v4* (2026).
- [32] J. Yu, Z. Feng, J. Lu, T. Cai, D. Zhou, Time-aware attention for enhanced electronic health records modeling, *arXiv:2507.14847v1* (2025).
- [33] M. Mesinovic, M. Buhlan, T. Zhu, Causal graph neural networks for healthcare, *arXiv:2511.02531v5* (2026).
- [34] Y. Zhang, S. Li, Learning clinical representations under systematic distribution shift, *arXiv:2603.07348v1* (2026).
- [35] R. Liu, I. Q. Mohiuddin, A. J. Schoeffler, K. Renduchintala, A. Nayak, P. L. Vemu, S. C. Vedak, K. C. Black, J. L. Havlik, I. Ogunmola, S. P. Ma, R. Dhatt, J. H. Chen, Physicianbench: Evaluating llm agents in real-world ehr environments, *arXiv:2605.02240v1* (2026).
- [36] H. Chen, D. Metelski, L. Qi, T. Xia, J. Lee, S. Brown, K. Riley, F. Wang, H. Capps, Z. Tang, X. Song, L. Kong, F. Feng, T. Zeng, Z. Liu, Z. Ma, H. Jiang, F. Geng, Y. Yuan, C. You, Q. Wen, H. Wei, Y. Fu, Y. Zhao, C. Yang, B. Huang, K. Zhang, C. Xiong, S. Koyejo, E. P. Xing, P. S. Yu, W. Yao, Chi-bench: Can ai agents

automate end-to-end, long-horizon, policy-rich healthcare workflows?, arXiv:2605.16679v2 (2026).

- [37] H. Zhang, N. Dullerud, L. Seyyed-Kalantari, Q. Morris, S. Joshi, M. Ghassemi, An empirical framework for domain generalization in clinical settings (2021).
- [38] V. Manthena, of Alberta, of Alberta, Federated learning and differential privacy techniques on multi-hospital population-scale electrocardiogram data (2024).
- [39] D. Liu, Scaffolded turns and logical conversations: Designing humanized llm-powered conversational agents for hospital admission interviews (2025).
- [40] E. G. Woo, Synthetic data distillation enables the extraction of clinical information at scale, npj Digital Medicine (2025).
- [41] H. SHARMA, Causal representation learning for predicting autoimmune disease progression from longitudinal multimodal clinical data (2025).
- [42] M. Jayasurya, A. information, Federated learning for privacy-preserving healthcare data analytics, nature (2026).
- [43] A. Ghosh, T. Ashraf, R. K. Singh, N. Saeed, S. Saha, X. Chen, S. Khan, Carepilot: A multi-agent framework for long-horizon computer task automation in healthcare.
- [44] A. M. Vahdani, M. Shariatnia, P. Rajpurkar, A. Pareek, Towards trustworthy artificial intelligence in musculoskeletal medicine: A narrative review on uncertainty quantification (2025).
- [45] C. S. Lee, Clinical applications of continual learning machine learning, The Lancet Digital Health (2020).
- [46] L. T. B. Hons, Medical artificial intelligence for clinicians: the lost cognitive perspective, The Lancet Digital Health (2024).
- [47] K. Thorlund, A real-time dashboard of clinical trials for covid-19, The Lancet Digital Health (2020).

- [48] B. Mao, Assessing risk factors for sars-cov-2 infection in patients presenting with symptoms in shanghai, china: a multicentre, observational cohort study, *The Lancet Digital Health* (2020).
- [49] G. Chevance, Digital health at the age of the anthropocene, *The Lancet Digital Health* (2020).
- [50] L. H. Schwamm, Virtual care: new models of caring for our patients and workforce, *The Lancet Digital Health* (2020).
- [51] P. B. A. Quarta, F. Calimeri, A. N. P. online January, Continual learning in medicine: A systematic literature (2025).
- [52] P. Zhou, J. Pujara, X. Ren, X. Chen, H.-T. Cheng, Q. V. Le, E. H. Chi, D. Zhou, S. Mishra, H. S. Zheng, Self-discover: Large language models self-compose reasoning structures, in: *Proceedings of the*, 2022.
- [53] H. Yao, R. Zhang, J. Huang, J. Zhang, Y. Wang, B. Fang, R. Zhu, Y. Jing, S. Liu, G. Li, D. Tao, A survey on agentic multimodal large language models, *arXiv:2510.10991v1* (2025).
- [54] B. Urooj, M. Fayaz, S. Ali, K. W. Kim, Large language models in medical image analysis: A systematic survey and future directions (2025).
- [55] X. Chen, J. Xiang, S. Lu, Y. Liu, D. Shi, Evaluating large language models and agents in healthcare: key challenges in clinical applications (2025).
- [56] Z. A. Nazi, W. Peng, Large language models in healthcare and medical domain: A review (2024).
- [57] J. C. L. Chow, K. Li, Large language models in medical chatbots: Opportunities, challenges, and the need to address ai risks (2025).
- [58] R. AlSaad, A. Abd-alrazaq, S. Boughorbel, A. Ahmed, M.-A. Renault, R. Damseh, J. Sheikh, Multimodal large language models in health care: Applications, challenges, and future outlook, *Journal of Medical Internet Research* 26 (2024) e59505. doi:10.2196/59505.
URL <https://www.jmir.org/2024/1/e59505>

- [59] R. Nagarajan, M. Kondo, F. Salas, E. Sezgin, Y. Yao, V. Klotzman, S. A. Godambe, N. Khan, A. Limon, G. Stephenson, S. Taraman, N. Walton, L. Ehwerhemuepha, J. Pandit, D. Pandita, M. Weiss, C. Golden, A. Gold, J. Henderson, A. Shippy, L. A. Celi, W. R. Hogan, E. K. Oermann, T. Sanger, S. Martel, Economics and equity of large language models: Health care perspective, *Journal of Medical Internet Research* 26 (2024) e64226. doi:10.2196/64226.
URL <https://www.jmir.org/2024/1/e64226>
- [60] K. Zhang, X. Meng, X. Yan, J. Ji, J. Liu, H. Xu, H. Zhang, D. Liu, J. Wang, X. Wang, J. Gao, Y. geng-shuo Wang, C. Shao, W. Wang, J. Li, M.-Q. Zheng, Y. Yang, Y.-D. Tang, Revolutionizing health care: The transformative impact of large language models in medicine, *Journal of Medical Internet Research* 27 (2025) e59069. doi:10.2196/59069.
URL <https://www.jmir.org/2025/1/e59069>
- [61] X. Yang, Y. Xiao, D. Liu, H. Deng, J. Huang, M. Liang, L. Dong, Z. Yuan, J. Yao, W. Guo, C. Xu, X. Yang, Z. E. Road, Factors influencing adoption of large language models in health care: Multicenter cross-sectional mixed methods observational study (2025).
- [62] D. McDuff, Towards accurate differential diagnosis with large language models, *Nature* (2025).
- [63] N. Mehandru, Evaluating large language models as agents in the clinic, *npj Digital Medicine* (2024).
- [64] J. Zhou, Large language models in biomedicine and healthcare (2025).
- [65] S. Zhou, Explainable differential diagnosis with dual-inference large language models (2025).
- [66] J. Logan, Continuum memory architectures for long-horizon llm agents., *arXiv* (2026).
- [67] A. Agweyu, Generative ai-enabled clinical decision support system in primary care: a pragmatic, cluster-randomized trial, *Nature Medicine* (2025).
- [68] H. Ying, Y. Lia, Z. Fu, Domain adaptation and generalization using foundation models in healthcare imaging.

- [69] W. Chen, F. Zhang, M. Li, X. Wang, Z. Liu, R. Huang, J. Zhao, Y. Lia, Domain shift and generalization strategies for foundation models in clinical imaging (2025).
- [70] R. J. Limaye, Building trust while influencing online covid-19 content in the social media world, *The Lancet Digital Health* (2020).
- [71] A. M. Association, Addressing distribution shift for robust and trustworthy prediction and causal inference in clinical ai settings, *JAMA* (2025).
- [72] S. A. C. Ordóñez, Beyond overconfidence: Embedding curiosity and humility for ethical medical ai (2026).
- [73] H. Haick, R. A. R. A. A. May, Federated learning in multimodal healthcare diagnostics: Privacy- preserving ai for biomedical imaging, electronic health records, wearables, and clinical decision support (2026).
- [74] S. C. Nouis, V. Uren, S. Jariwala, Evaluating accountability, transparency, and bias in ai-assisted healthcare decision- making: a qualitative study of healthcare professionals’ perspectives in the uk (2025).
- [75] A. van den Oord, Y. Li, O. Vinyals, Representation learning with contrastive predictive coding., *arXiv* (2018).
- [76] X. Hong, Y. Shi, Y. Li, Y.-W. Chen, Alzheimers disease progression prediction based on manifold mapping of irregularly sampled longitudinal data, *arXiv:2511.20154v1* (2025).
- [77] H. M. Papachappa, Semantic intent weaver: A context-aware search algorithm for ambiguous query resolution., *ICCIDS* (2026).
- [78] S. Bakas, X. Li, P. Shah, B. Columbia, Federated learning in healthcare: From research to real-world deployment (2026).
- [79] e. a. Chen, T., A simple framework for contrastive learning of visual representations., in: *ICML*, 2020.
- [80] I. Forecasting, H. M. Papachappa, A. Kumar, N. Badam, Riemannian intent manifolds for session-based (2019).

- [81] A. E. W. Johnson, T. J. Pollard, L. Shen, L.-w. H. Lehman, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L. A. Celi, R. G. Mark, Mimic-iii, a freely accessible critical care database, *Scientific Data* 3 (2016) 160035. doi:10.1038/sdata.2016.35.
- [82] A. E. W. Johnson, L. Bulgarelli, L. Shen, A. Gayles, A. Shammout, S. Horng, T. J. Pollard, S. Hao, B. Moody, B. Gow, L.-w. H. Lehman, L. A. Celi, R. G. Mark, Mimic-iv, a freely accessible electronic health record dataset, *Scientific Data* 10 (1) (2023) 1. doi:10.1038/s41597-022-01899-x.
- [83] A. Johnson, T. Pollard, S. Horng, L. A. Celi, R. Mark, Mimic-iv-note: Deidentified free-text clinical notes (version 2.2) (2023). doi:10.13026/1n74-ne17.
URL <https://doi.org/10.13026/1n74-ne17>
- [84] A. E. W. Johnson, T. J. Pollard, S. J. Berkowitz, N. R. Greenbaum, M. P. Lungren, C.-y. Deng, R. G. Mark, S. Horng, Mimic-cxr, a de-identified publicly available database of chest radiographs with free-text reports, *Scientific Data* 6 (2019) 317. doi:10.1038/s41597-019-0322-0.
- [85] T. J. Pollard, A. E. W. Johnson, J. D. Raffa, L. A. Celi, R. G. Mark, O. Badawi, The eicu collaborative research database, a freely available multi-center database for critical care research, *Scientific Data* 5 (2018) 180178. doi:10.1038/sdata.2018.178.
- [86] M. Faltys, M. Zimmermann, X. Lyu, M. Hüser, S. Hyland, G. Rätsch, T. Merz, Hirid, a high time-resolution icu dataset (version 1.1.1) (2021).
- [87] G. Salton, M. J. McGill, *Introduction to Modern Information Retrieval*, McGraw-Hill, 1983.
- [88] S. E. Robertson, S. Walker, S. Jones, M. M. Hancock-Beaulieu, M. Gatford, Okapi at trec-3, in: *Proceedings of the Third Text REtrieval Conference (TREC-3)*, 1995.
- [89] S. E. Robertson, K. Sparck Jones, Relevance weighting of search terms, *Journal of the American Society for Information Science* 27 (3) (1976) 129–146.

- [90] I. U. Haq, M. Mhamed, M. Al-Harbi, H. Osman, Z. Y. Ha, Z. Liu, Advancements in medical radiology through multimodal machine learning: A comprehensive overview (2025).
- [91] B. Jandoubi, M. A. Akhloufi, Multimodal artificial intelligence in medical diagnostics (2025).
- [92] U. Shanmugam, M. K. Rajendran, J. Natarajan, V. V. S. R. Karri, Clinical trial design and regulatory requirements for artificial intelligence as a medical device: A prisma-scr-guided scoping review of global guidance and evidence (2017–2025) (2026).
- [93] M. Hoche, What makes clinical machine learning fair? a practical ethics framework (2025).
- [94] M. Sorka, A multi-agent approach to neurological clinical reasoning (2025).
- [95] R. Johnson, G. Shaham, L. Eisen, J. Waxman, S. Devons, C. R. Ginder, P. Hong, R. Sayeed, B. Y. Reis, N. Dagan, H. Services, Clinvec: Unified embeddings of clinical codes enable knowledge-grounded ai in medicine (2025).
- [96] F. Fontin, J. Honerlaw, M. Murray, A. Galloway, D. Heise, K. Connatser, L. Davies, J. Gosian, M. Maripuri, J. Russo, R. Sangar, V. Tanukonda, E. Zielinski, M. Dubreuil, A. J. Zimolzak, Centralized interactive phenomics resource: an integrated online phenomics knowledgebase for health data users, *Journal of the American Medical Informatics Association* (2024).
- [97] M. Arora, H. Mortagy, N. Dwarshuis, J. Wang, A. L. Holder, S. Gupta, R. Kamaleswaran, P. Yang, Improving clinical decision support through interpretable machine learning and error handling in electronic health records, *Journal of the American Medical Informatics Association* (2025).
- [98] P. Sanchez, J. P. Voisey, T. Xia, A. Q. O’Neil, S. A. Tsiftaris, V. JP, O. AQ, T. S. Causal, R. May, A. July, S. Category, C. science, S. Areas, Causal machine learning for healthcare and precision (2022).

- [99] M. Perkonigg, Dynamic memory to alleviate catastrophic forgetting in continual learning with medical imaging, *Nature* (2021).
- [100] E. Klang, Orchestrated multi agents sustain accuracy under clinical-scale workloads compared to a single agent (2026).
- [101] X. Li, Y. Chen, Y. Duan, S. Wu, A. Yu, C.-H. Chiu, J. Ni, G. Xu, T. J.-J. Li, A. Yuille, W. Jin, Y. Shi, N. Tang, A comprehensive survey of agentic ai in healthcare (2025).
- [102] Y. Wang, A. Ostropelets, J. J. Mulgrave, D. M. Blei, G. Hripcsak, The medical deconfounder: Assessing treatment effects with electronic health records linying zhang, in: *Proceedings of Machine*, 2018.
- [103] G. Cicceri, Dilocc: An approach for distributed incremental learning across the computing continuum (2021).
- [104] G. Romeo, Exploring automation bias in human–ai collaboration: a review and implications for explainable ai (2015).
- [105] F. Hasanzadeh, Bias recognition and mitigation strategies in artificial intelligence healthcare applications, *npj Digital Medicine* (1993).
- [106] F. Chen, L. Wang, J. Hong, J. Jiang, L. Zhou, Unmasking bias in artificial intelligence: a systematic review of bias detection and mitigation strategies in electronic health record-based models, *Journal of the American Medical Informatics Association* (2024).
- [107] J. Amann, Explainability for artificial intelligence in healthcare: a multidisciplinary perspective, *BMC Medical Informatics and Decision Making* (2020).
- [108] K. Lekadir, B. Glocker, C. Cintas, C. P. Langlotz, E. Weicken, F. Prior, G. Kaissis, G. Tsakou, J. Kalpathy-Cramer, J. Mongan, K. Kushibar, K. Riklund, K. Marias, A. training, Future-ai: international consensus guideline for trustworthy and deployable artificial intelligence in healthcare, *BMJ* (2025).
- [109] D. Moher, A. Liberati, J. Tetzlaff, D. G. Altman, Preferred reporting items for systematic reviews and meta-analyses: the prisma statement, *bmj* (2009).

- [110] J. Li, Prioritizing human-ai collaboration in healthcare: the triad framework for trustworthy governance, real-world, and integrated adaptive deployment (2025).